Lab6

Packet Tracer - WLAN Configuration

Addressing Table

Device	Interface	IP Address
Home Wireless Router	Internet	DHCP
	LAN	192.168.6.1/27
RTR-1	G0/0/0.2	192.168.2.1/24
	G0/0/0.5	192.168.5.1/24
	G0/0/0.100	192.168.100.1/24
	G0/0/1	10.6.0.1/24
SW1	VLAN 200	192.168.100.100/24
LAP-1	G0	DHCP
WLC-1	Management	192.168.100.254/24
RADIUS Server	NIC	10.6.0.254/24
Home Admin	NIC	DHCP
Enterprise Admin	NIC	192.168.100.200/24
Web Server	NIC	203.0.113.78/24
DNS Server	NIC	10.100.100.252
Laptop	NIC	DHCP
Tablet PC	Wireless0	DHCP
Smartphone	Wireless0	DHCP
Wireless Host 1	Wireless0	DHCP
Wireless Host 2	Wireless0	DHCP

WLAN Information

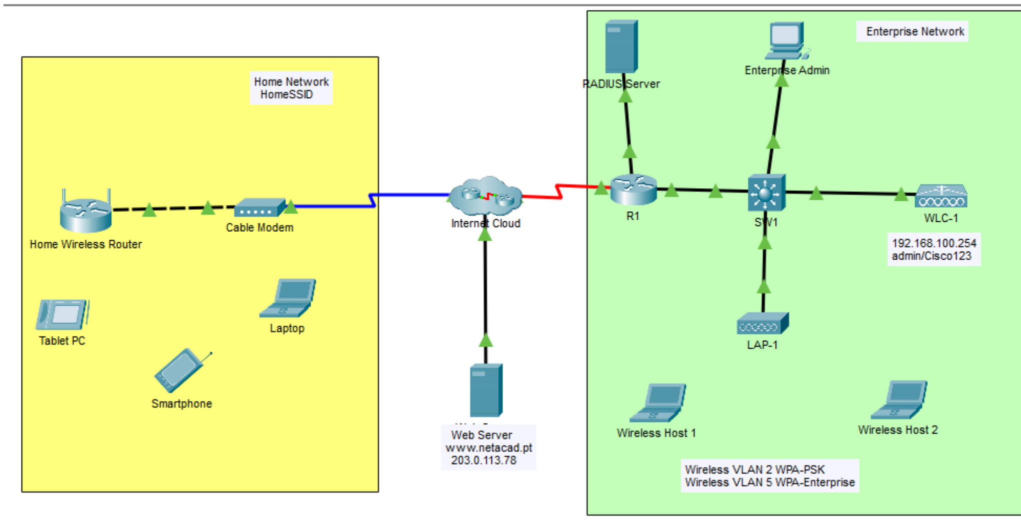
WLAN	SSID	Authentication	Username	Password
Home Network	HomeSSID	WPA2-Personal	N/A	Cisco123
WLAN VLAN 2	SSID-2	WPA-2 Personal	N/A	Cisco123
WLAN VLAN 5	SSID-5	WPA-2 Enterprise	userWLAN5	userW5pass

Note: It is not a good practice to reuse passwords as is done in this activity. Passwords have been reused to make it easier to work through the tasks.

Objectives

In this activity, you will configure both a wireless home router and a WLC-based network. You will implement both WPA2-PSK and WPA2-Enterprise security.

- Configure a home router to provide Wi-Fi connectivity to a variety of devices.
- Configure WPA2-PSK security on a home router.
- Configure interfaces on a WLC.
- Configure WLANs on a WLC.
- Configure WPA2-PSK security on a WLAN and connect hosts to WLAN.
- Configure WPA2-Enteprise on a WLAN and connect hosts to the WLAN.
- Verify connectivity WLAN connectivity.



Background / Scenario

You will apply your WLAN skills and knowledge by configuring a home wireless router and an enterprise WLC. You will implement both WPA2-PSK and WPA2-Enterprise security. Finally, you will connect hosts to each WLAN and verify connectivity.

Instructions

Part 1: Configure a Home Wireless Router.

You are installing a new home wireless router at a friend's house. You will need to change settings on the router to enhance security and meet your friend's requirements.

Step 1: Change DHCP settings.

a. Open the Home Wireless Router GUI and change the router IP and DHCP settings according to the information in the Addressing Table.

b. Permit a maximum of **20** addresses to be issued by the router.

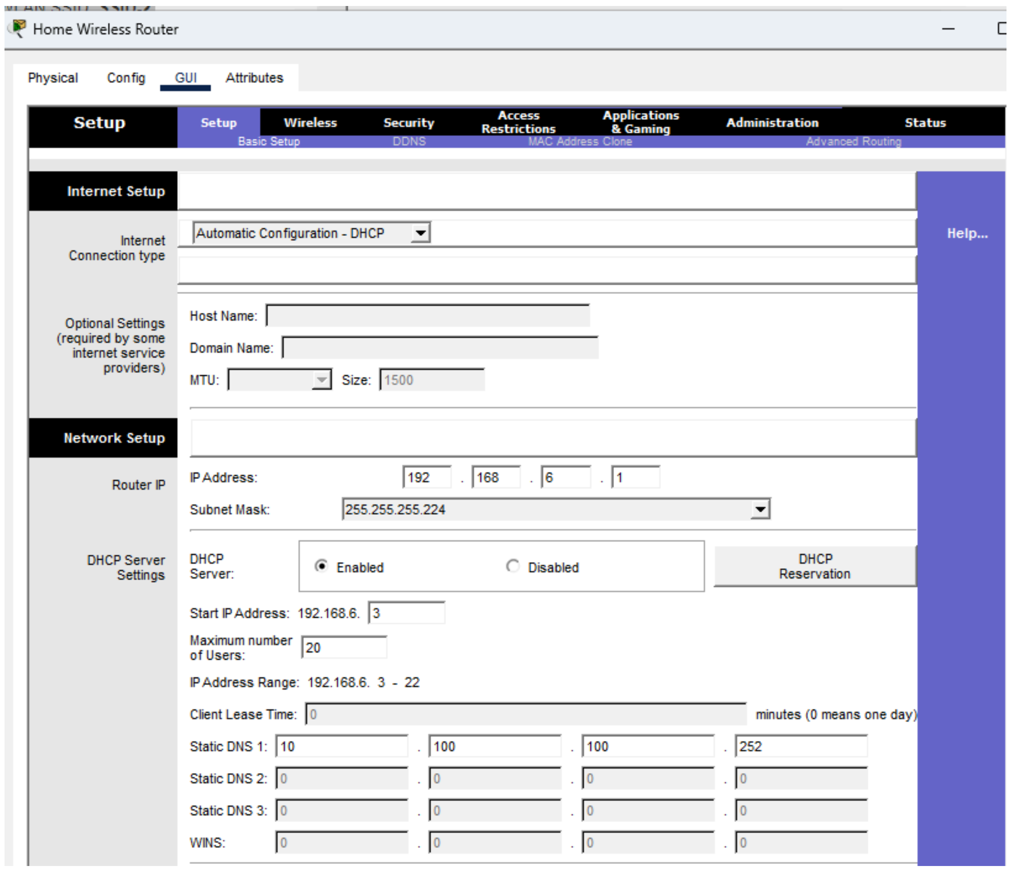
c. Configure the DHCP server to start with IP address .**3** of the LAN network.

d. Configure the internet interface of the router to receive its IP address over DHCP.

Verify the address. What address did it receive?

It's 192.168.6.1

e. Configure the static DNS server to the address in the Addressing Table.

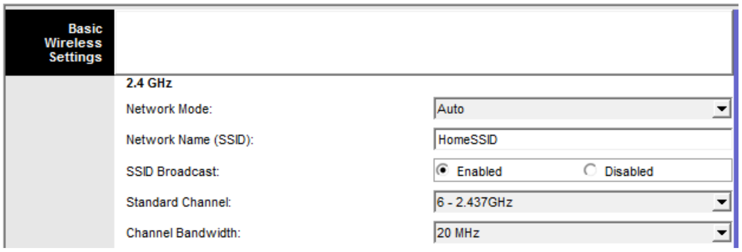


Step 2: Configure the Wireless LAN.

a. The network will use the 2.4GHz Wireless LAN interface. Configure the interface with the SSID shown in the Wireless LAN information table.

b. Use **channel 6**.

c. Be sure that all wireless hosts in the home will be able to see the SSID.

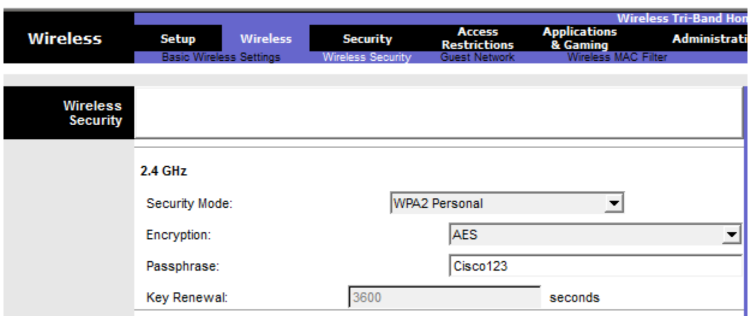


Make it use 2.4GHz Fill “HomeSSID” on the Network Name and use channel 6 as the tasks.

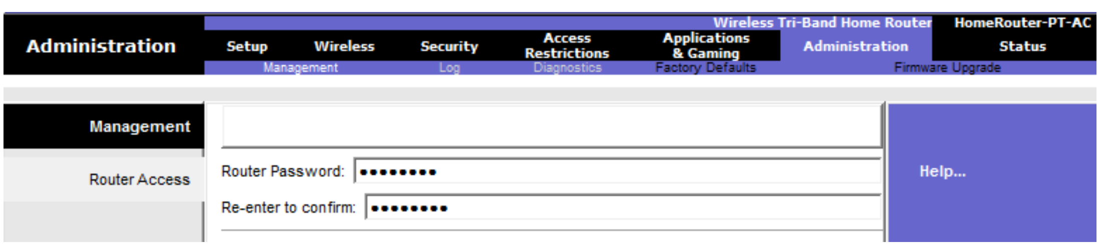
Step 3: Configure security.

a. Configure wireless LAN security. Use WPA2 Personal and the passphrase shown in the Wireless LAN information table.

b. Secure the router by changing the default password to the value shown in the Wireless LAN information table.



Setting security Mode to WPA2 Personal and Passphrase to “Cisco123”

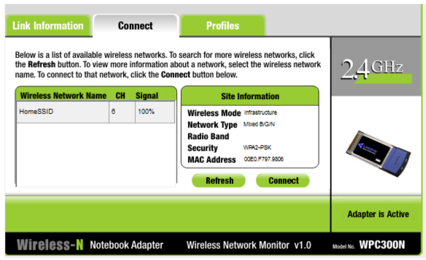


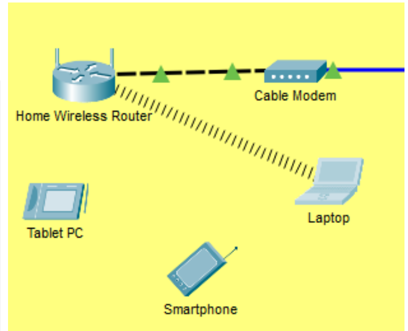
Changing Admin password of the router to “Cisco123”

Step 4: Connect clients to the network.

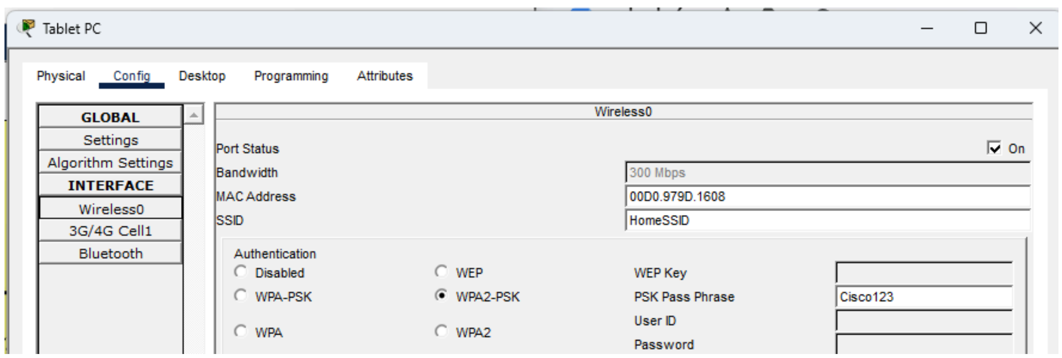
a. Open the PC Wireless app on the desktop of the laptop and configure the client to connect to the network.

b. Open the Config tab on the Tablet PC and Smartphone and configure the wireless interfaces to connect to the wireless network.

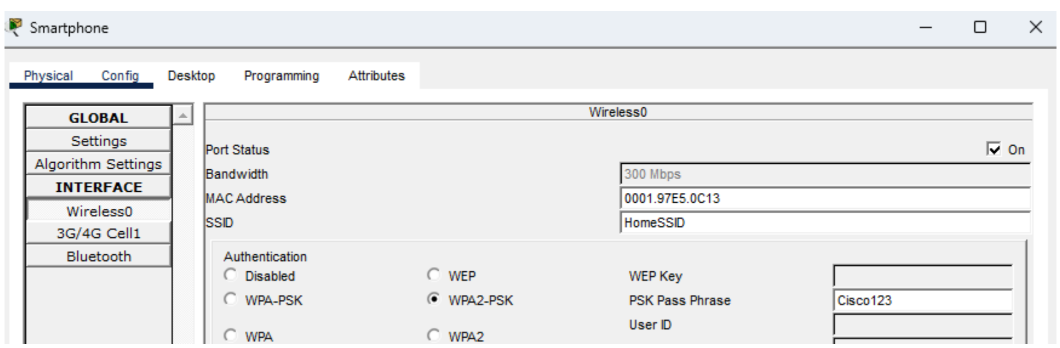




On the Laptop: IPv4 Address 192.168.6.3

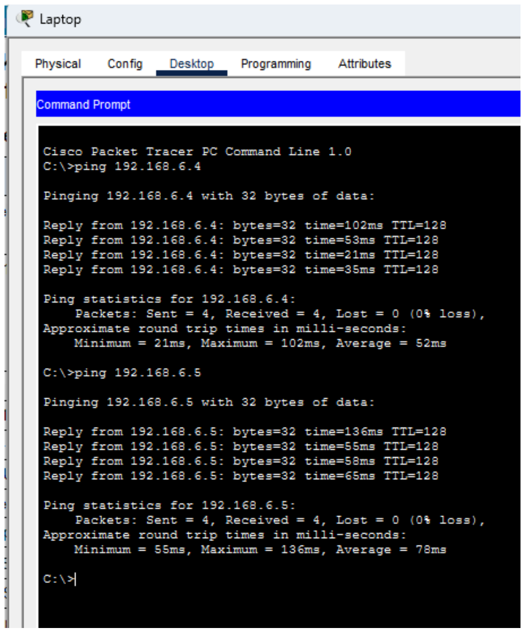


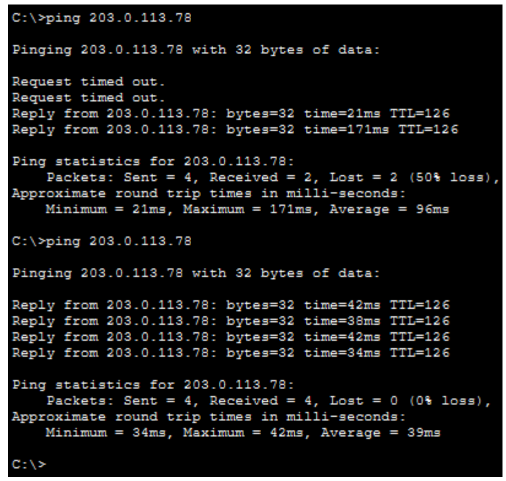
For the Tablet PC. Fill the SSID box with “Home SSID” and select Authentication to WPA-PSK and PSK Pass Phrase to “Cisco123”



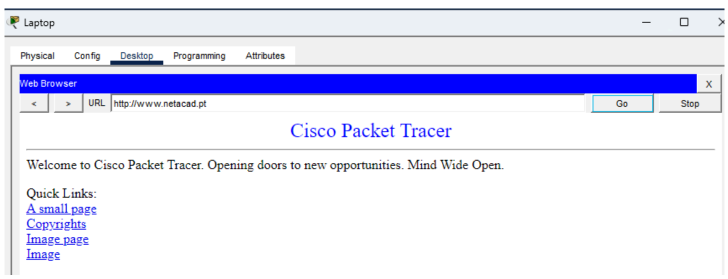
And then, for the Smartphone. Fill the SSID box with “Home SSID” and select Authentication to WPA-PSK and PSK Pass Phrase to “Cisco123”

c. Verify connectivity. The hosts should be able to ping each other and the web server. They should also be able to reach the web server URL.





From these pictures you can see that they are pinging the other device successfully, but not for the Webserver. The Webserver pinging needed more time and in the first time it loss 50%.



Reach the web server by using the URL on Web Browser.

Part 2: Configure a WLC Controller Network

Configure the wireless LAN controller with two WLANs. One WLAN will use WPA2-PSK authentication. The other WLAN will use WPA2-Enterprise authentication. You will also configure the WLC to use an SNMP server and configure a DHCP scope that will be used by the wireless management network.

Step 1: Configure VLAN interfaces.

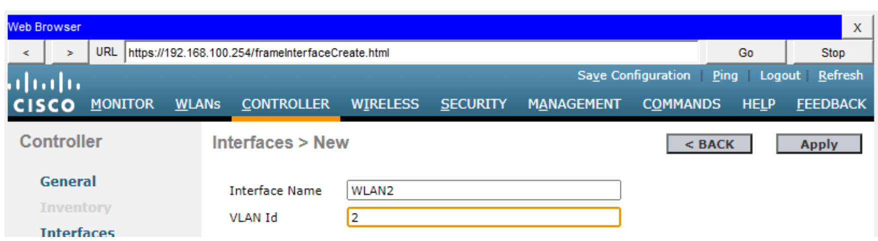
a. From the **Enterprise Admin**, navigate to the WLC-1 management interface via a web browser.

To log into WLC-1, use **admin** as the username and **Cisco123** as the password.

b. Configure an interface for the first WLAN.

Name: **WLAN 2**

VLAN Identifier: **2**



Port Number: **1**

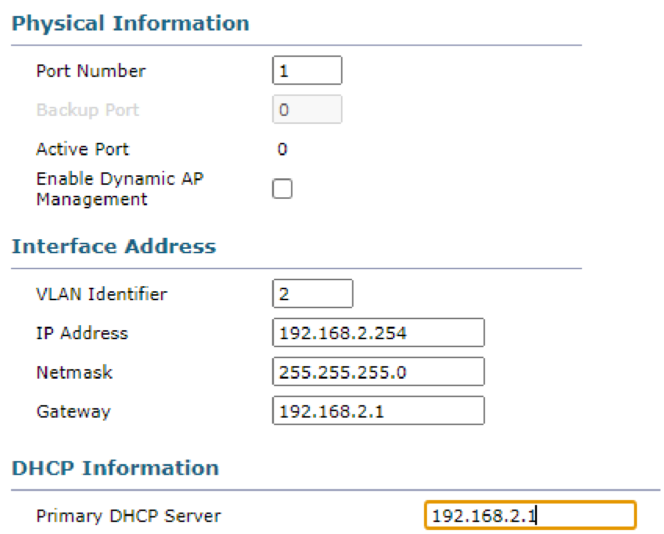
Interface IP Address: **192.168.2.254**

Netmask: **255.255.255.0**

Gateway: **RTR-1 G0/0/0.2 address**

Primary DHCP Server: **Gateway address**

c. Configure an interface for the second WLAN.



Name: **WLAN 5**

VLAN Identifier: **5**

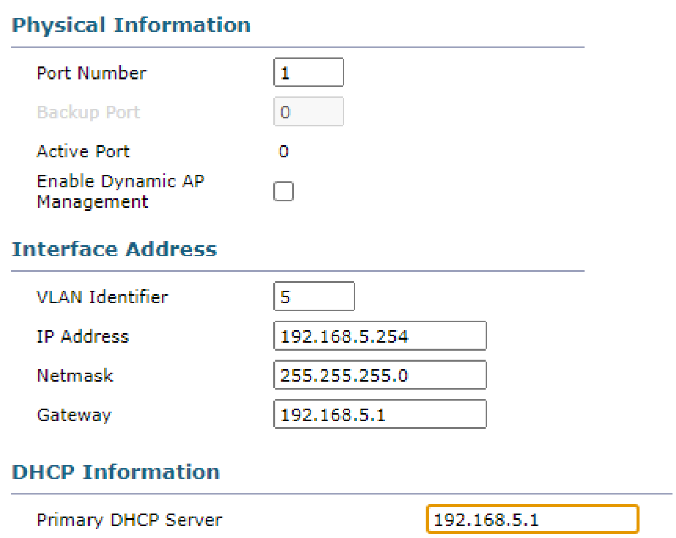
Port Number: **1**

Interface IP Address: **192.168.5.254**

Netmask: **255.255.255.0**

Gateway: **RTR-1 interface G0/0/0.5 address**

Primary DHCP Server: **Gateway address**



This is a WLAN that has already been created.

Interface Name	VLAN Identifier	IP Address	Interface Type	Dynamic AP Management	IPv6 Address	
WLAN2	2	192.168.2.254	Dynamic	Disabled		Remove
WLAN5	5	192.168.5.254	Dynamic	Disabled		Remove
management	1	192.168.100.254	Static	Enabled	::/128	
virtual	N/A	192.0.2.1	Static	Not Supported		

WLAN2 and WLAN5 are created.

Step 2: Configure a DHCP scope for the wireless management network.

Configure and enable an internal DHCP scope as follows:

Scope Name: **management**

Pool Start Address: **192.168.100.235**

Pool End Address: **192.168.100.245**

Network: **192.168.100.0**

Netmask: **255.255.255.0**

Default Routers: **192.168.100.1**

Scope Name	Address Pool	Lease Time
management	192.168.100.235 - 192.168.100.245	

Step 3: Configure the WLC with external server addresses.

a. Configure the RADIUS server information as follows:

Sever Index: **1**

Server Address: **10.6.0.254**

Shared Secret: **RadiusPW**

Network User	Management	Server Index	Server Address(Ipv4/Ipv6)	Port	IPSec	Admin Status	
☑	☑	1	10.6.0.254	1812	Disabled	Enabled	Remove

After configuring the Radius Server, now the Admin Status is Enabled

b. Configure the WLC to send logs information to an SNMP server.

Community Name: **WLAN**

IP Address: **10.6.0.254**

SNMP v1 / v2c Community

Apply New...

Community Name	IP Address(Ipv4/Ipv6)	IP Mask/Prefix Length	Access Mode	Status
public	0.0.0.0	0.0.0.0	Read-Only	Enable
private	0.0.0.0	0.0.0.0	Read-Write	Enable

IPSec Parameters

IPSec ☐

IPSec Profile Name: none

Seems like the New button in Management is disabled to use… A question for TA: How can I make it enable?

Step 4: Create the WLANs.

a. Create the first WLAN:

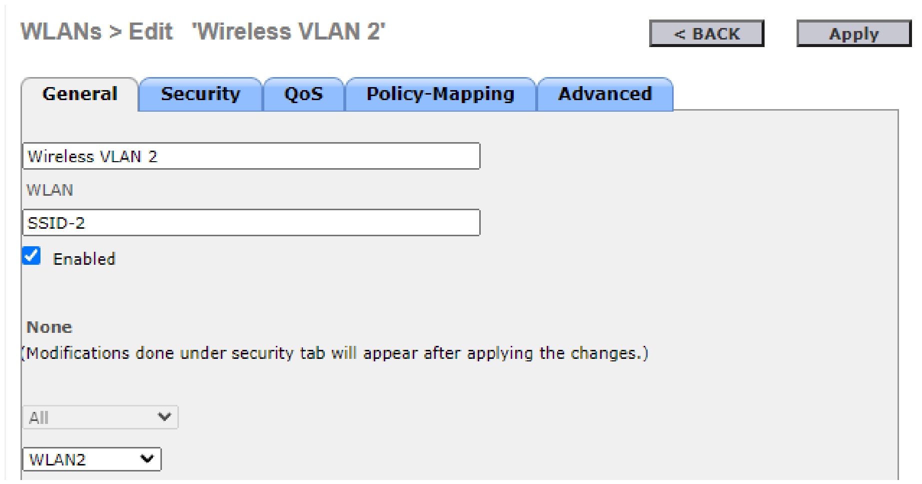
Profile Name: **Wireless VLAN 2**

WLAN SSID: **SSID-2**

ID: **2**

Interface: **WLAN 2**

After creating it. It will navigate to the Edit 'Wireless VLAN 2' that is the picture below.



Security: **WPA2-PSK**

Passphrase: **Cisco123**

Under the Advanced tab, go to the FlexConnect section. Enable **FlexConnect Local Switching** and **FlexConnect Local Auth**.

b. Create the second WLAN:

Profile Name: **Wireless VLAN 5**

WLAN SSID: **SSID-5**

Interface: **WLAN 5**

ID: **5**

Security: **802.1x - WPA2-Enterprise**

Configure the WLAN to use the RADIUS server for authentication.

Make the **FlexConnect** settings as was done in Step 4a.

WPA2-Enterprise = WPA2 (encryption) + 802.1X (authentication method via RADIUS) unlike WPA2-Personal which = WPA2 + PSK (fixed password).

Packet Tracer therefore allows us to select it by way of the "Auth Key Mgmt" field instead of having a direct "WPA2-Enterprise" dropdown.

WLANs

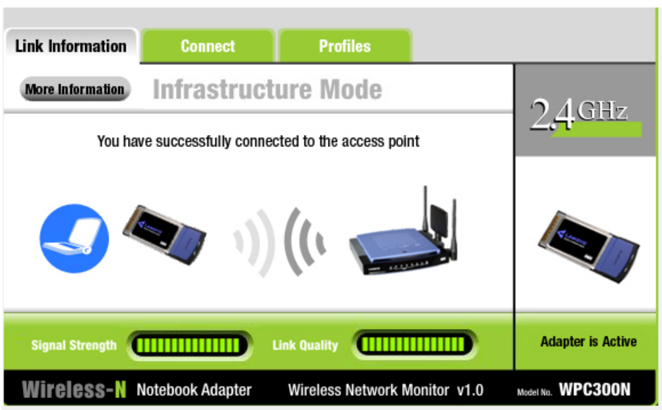
Current Filter: [Change Filter] [Clear Filter] Create New Go

WLAN ID	Type	Profile Name	WLAN SSID	Admin Status	Security Policies	
2	WLAN	Wireless VLAN 2	SSID-2	Enabled	[WPA2][Auth(PSK)]	Remove
5	WLAN	Wireless VLAN5	SSID-5	Enabled	[WPA2][Auth(802.1X)]	Remove

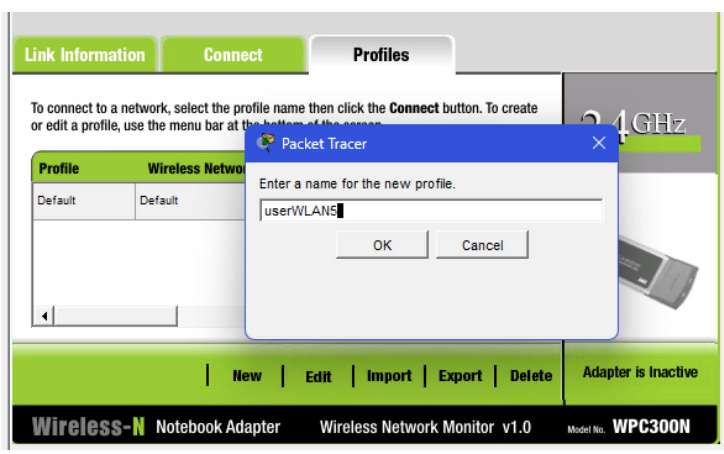
Step 5: Configure the hosts to connect to the WLANs.

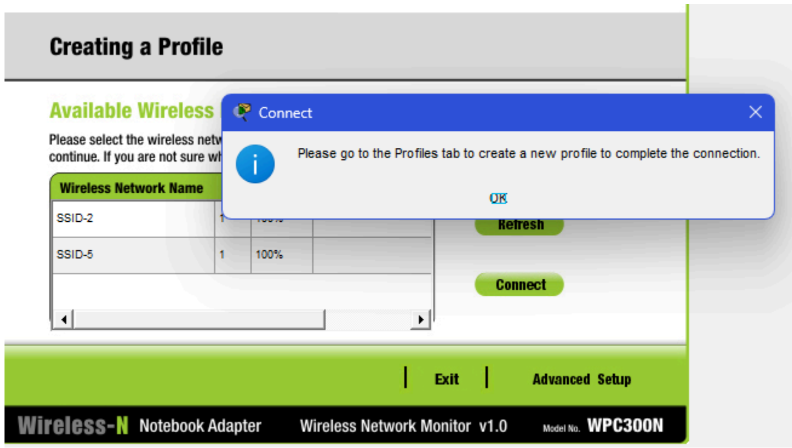
Use the desktop PC Wireless app to configure the hosts as follows:

a. Wireless Host 1 should connect to Wireless VLAN 2.



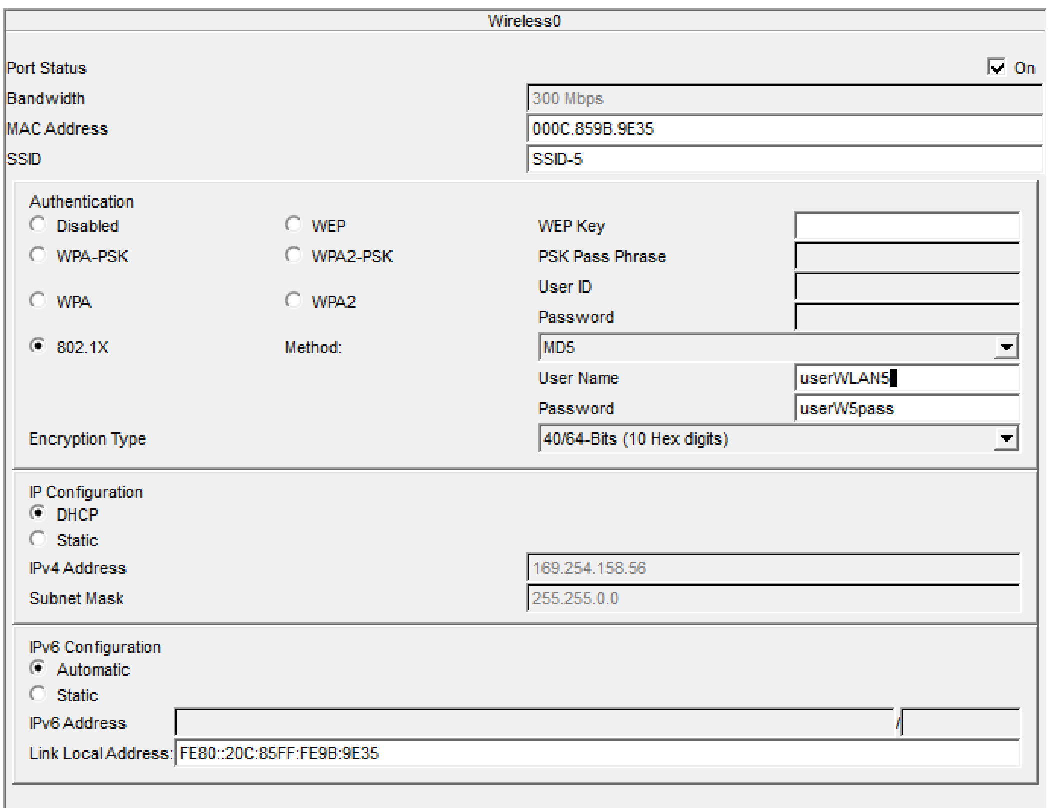
b. Wireless Host 2 should connect to Wireless VLAN 5 using the credentials in the WLAN information table.

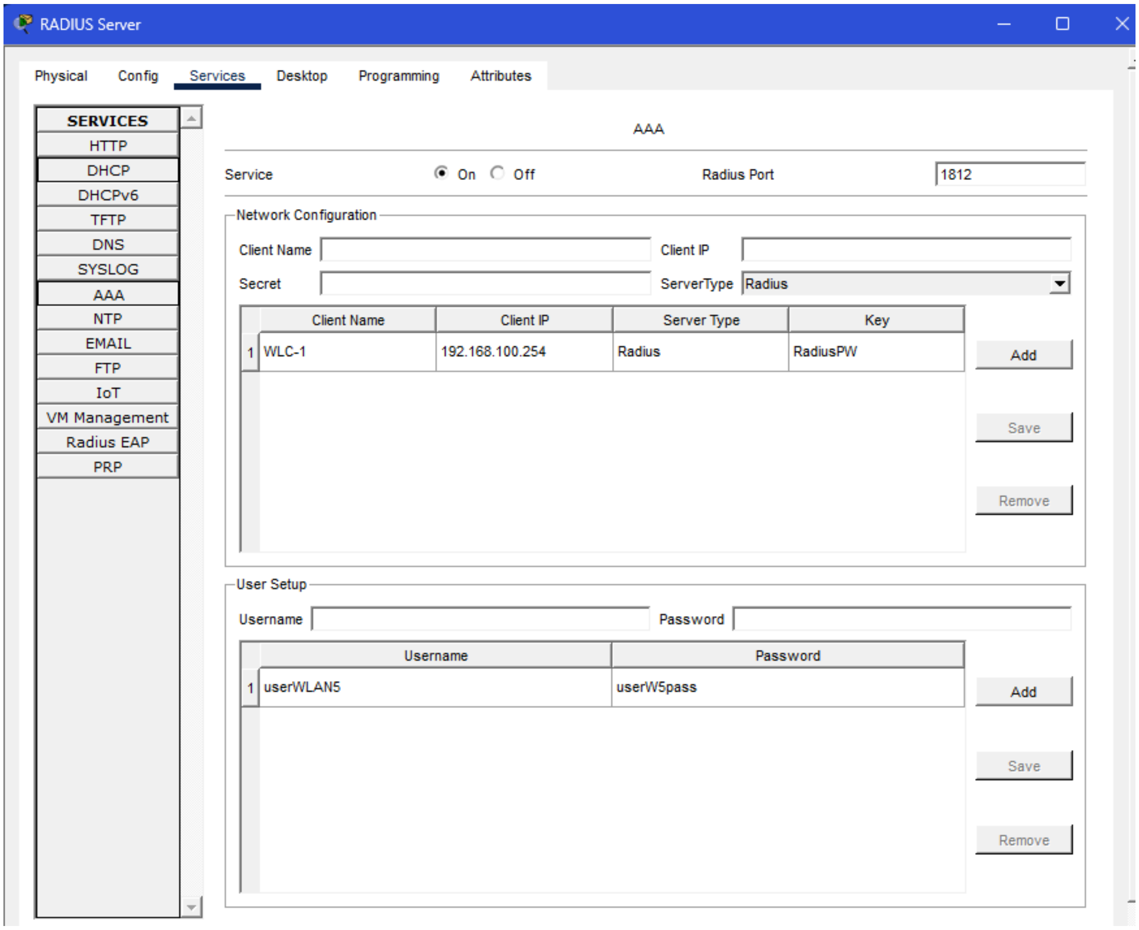


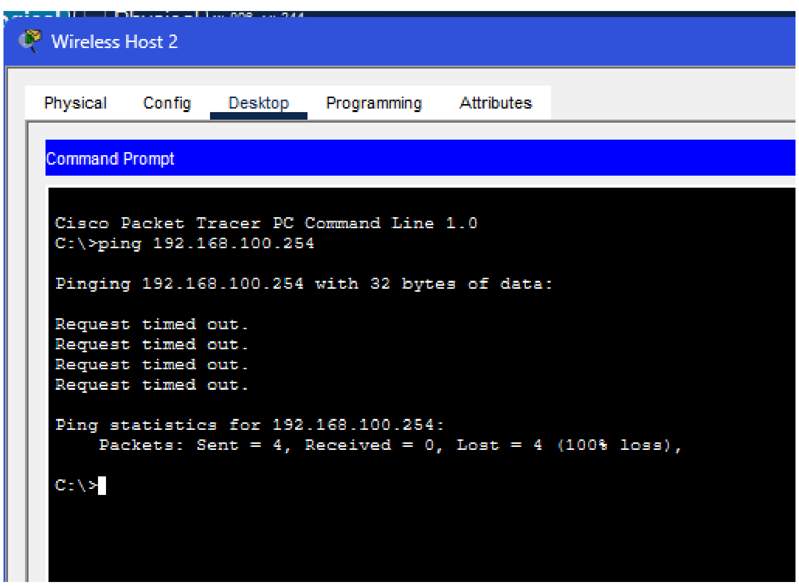


I'm in an endless loop.

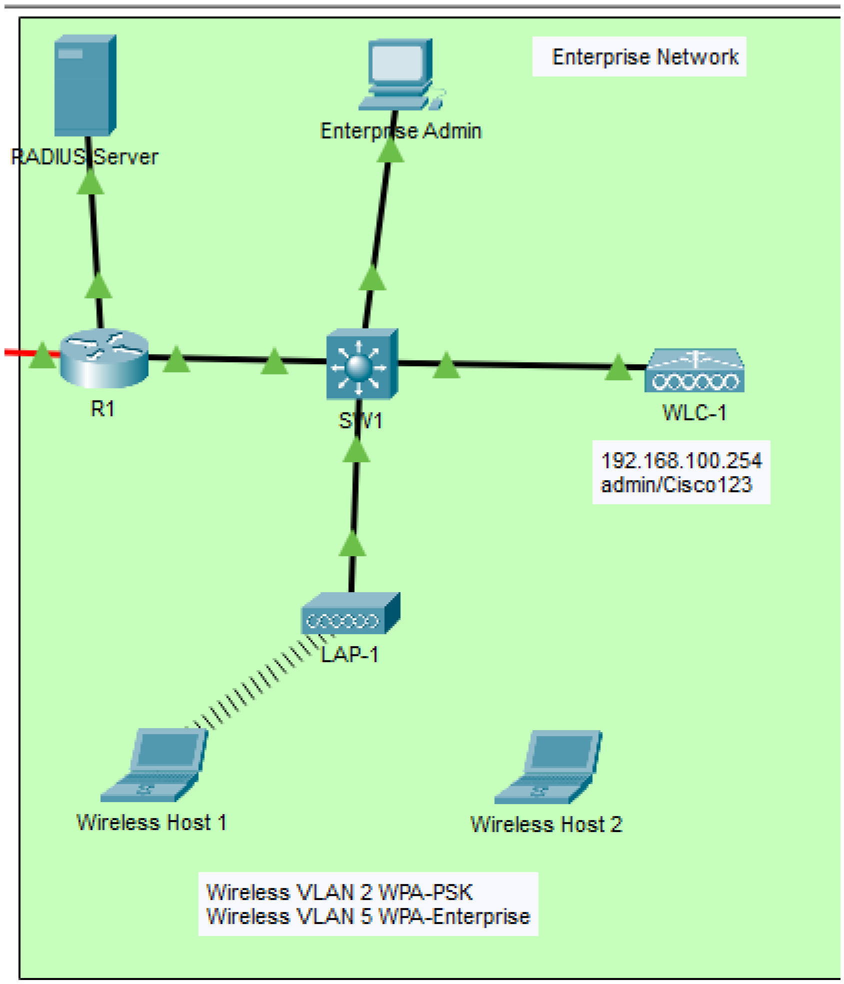
So I decided to use Config -> Interface -> Wireless0 instead for Wireless Host2.





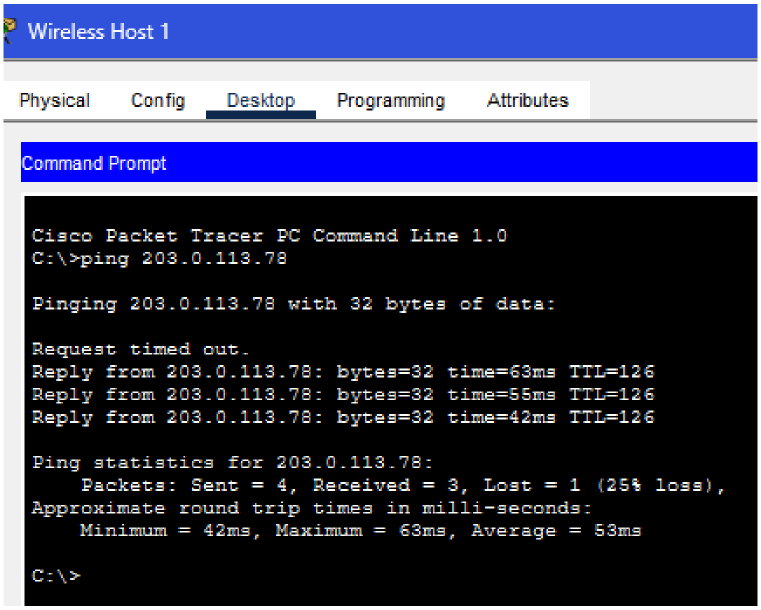


I cannot do this part. So this is the result of what I can do.



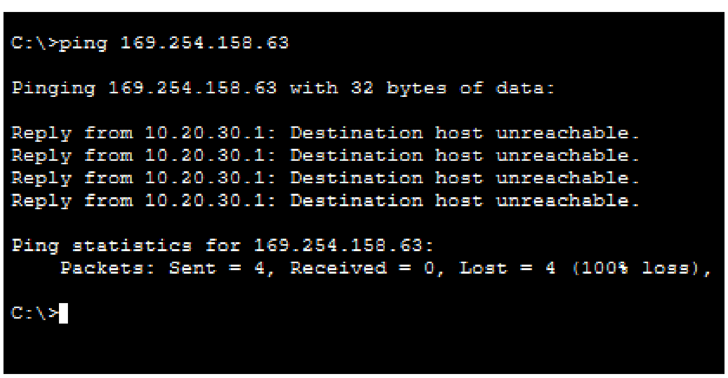
Step 6: Test connectivity.

Test connectivity between the wireless hosts and the Web Server by ping and URL.



Wireless Host 1 ping Web Server test.

Wireless Host 1 URL on **Web Browser to netcad**.



Wireless Host 1 ping to Wireless Host 2 and It fails.

WLAN 5 (Profile: Wireless VLAN 5, SSID: SSID-5) has been configured with

Layer 2 Security = WPA2, Auth Key Management = 802.1X, and is connected to the RADIUS Server (10.6.0.254).

A user account (userWLAN5) has also been created, and the connection between the WLC and the RADIUS Server has been confirmed to be working correctly (ping successful).

However, Packet Tracer encountered limitations in simulating 802.1X/EAP-MD5 authentication between the Wireless Host and the WLC, preventing successful testing of the actual connection of Wireless Host 2, even though all other settings were correct according to the specifications.